

Fleming's left-hand rule and motors

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 6 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	Describe the motor effect.	A current-carrying conductor in a magnetic field experiences a force.	☐
2 ★	In Fleming's left-hand rule, state what the thumb represents.	The direction of the force	☐
3 ★	In Fleming's left-hand rule, state what the first finger represents.	The direction of the magnetic field	☐
4 ★	In Fleming's left-hand rule, state what the second finger represents.	The direction of the current	☐
5 ★	Give three ways to increase the force on a current-carrying wire in a magnetic field.	Increase the magnetic flux density, the current, or the length of wire in the field.	☐
6	Define magnetic flux density.	A measure of the strength of a magnetic field.	☐
7	State the standard unit of magnetic flux density.	Tesla (T)	☐
8	Describe what happens to a current-carrying coil in a magnetic field.	It rotates, because the forces on its two sides act in opposite directions.	☐
9	Name the effect used in electric motors.	The motor effect	☐
10	State the equation for the force on a current-carrying conductor at right angles to a magnetic field.	force = magnetic flux density × current × length ($F = B I l$)	☐
11	Give two ways to reverse the force on a current-carrying wire in a magnetic field.	Reverse the current or reverse the magnetic field.	☐
12	Give two ways to make an electric motor spin faster.	Increase the current, or use a stronger magnetic field.	☐

	QUESTION	ANSWER	✓
13 ★	Explain why a current-carrying wire in a magnetic field experiences a force.	The magnetic field around the wire interacts with the field it is placed in.	☐